



University
Mohammed VI
Polytechnic



Deliverable #2: Relational Schema

Data Management Course

UM6P College of Computing

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Team Information

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Repository Link	://github.com/bsarah07/DatabasesCourseLabs

1 Introduction

The Research Lab Management System that we've created here is a database designed to manage a university by managing the multitude people, research activities, laboratories, equipment, reservations, and related resources of a research center. This deliverable focuses on transforming the RLMS conceptual ER model into a relational schema. The result identifies the relations, attributes, primary keys, foreign keys, and all what is required to represent the entities and relationships of the system while also making sure to respect the constraints defined in the conceptual design.

2 Requirements

This deliverable addresses the following requirements:

- Represent all entity sets and relationship sets of the ERD into relational tables.
- Include the attributes and primary key of each relation.
- Include foreign keys and the relations they reference.
- Correctly represent entities, one-to-many and many-to-many relationships, relationships with attributes, recursive relationships, ISA hierarchies, weak entities and identifying relationships, non-binary relationships, and aggregation.
- Include additional integrity constraints for requirements that cannot be represented using only primary and foreign keys.

3 Methodology

- For M-M relationships, we decided to create a separate table containing the keys of the participating entities, as is the case for the following relationships: Advises, Reports, Attaches, FundsLabs, FundsPrj, Requires, Holds, Reserves
- Incorporate the key of the 1-side as a foreign key in the N-side relation, as in the rest of the relationships.
- For recursive relationships, use a foreign key referencing the same relation.
- For weak entities, include the owner's primary key in the weak entity relation and make it part of the composite primary key.
- Represent ISA hierarchies using the subtype's primary key as a foreign key to its parent.

4 Implementation & Results

Include:

```

• CREATE DATABASE Lab2;
• USE Lab2;

• CREATE TABLE person (
    pid CHAR(12) PRIMARY KEY,
    firstName varchar(50),
    lastName varchar(50),
    email varchar(50),
    affiliation varchar(50),
    startDate date,
    endDate date
);
• CREATE TABLE Participates ( /*we opted for a table even though a person only has one proj
    personId char(12) NOT NULL,
    code INT,
    role ENUM('PI', 'Co-PI', 'Collaborator'),

    PRIMARY KEY (personId, code),

    FOREIGN KEY (personId) REFERENCES person(pid),

    FOREIGN KEY (code)
        REFERENCES ResearchProject(code)
);
#21: OK, 0 records retrieved in 21.925ms
#22: OK, 0 records retrieved in 9.023ms
#23: OK, 0 records retrieved in 22.998ms
#24: OK, 0 records retrieved in 23.586ms
#25: OK, 0 records retrieved in 26.377ms
#26: OK, 0 records retrieved in 25.2ms
#27: OK, 0 records retrieved in 22.78ms
#28: OK, 0 records retrieved in 15.017ms
#29: OK, 0 records retrieved in 11.325ms
#30: OK, 0 records retrieved in 8.602ms
#31: OK, 0 records retrieved in 24.784ms
#32: OK, 0 records retrieved in 27.593ms
#33: OK, 0 records retrieved in 24.951ms
/ SQL*Script execution completed in 1.775s. 33 statements executed successfully.

```

5 Discussion

- We had trouble knowing how to represent a M-M (at least one to many) relationship because how can we guarantee that the relation includes all students? As is the case with "Advises" relationship.
- Understanding weak entities: maintenance was particularly challenging because we had to understand why the equipment unit's key becomes part of the Maintenance primary key, resulting in (*Serial_number*, *start_timestamp*).
- We learned how to code in SQL and transform an ER diagram into tables.
- We also learned how to incorporate aggregation into our relations.

6 Conclusion

This lab allowed us to transform the ERD in the previous lab into a relational schema while preserving the main entities, relationships, and constraints of the system. We also applied different representation methods for the 1-M and M-M and recursive relationships, as well as ISA hierarchies, and weak entities. Through this process, we needed to identify primary keys and foreign keys while maintaining the integrity constraints.